

Proiect testare 1

Despre Docker

Exemple comenzi

Rulare comenzi

Despre Docker

Docker este o platformă open-source care permite dezvoltatorilor să creeze, să distribuie și să ruleze aplicații în containere – unități izolate care includ toate dependențele necesare pentru funcționare. Spre deosebire de mașinile virtuale, containerele Docker sunt mai rapide, mai ușoare și rulează direct pe sistemul de operare al gazdei, folosind resurse minime. Cu Docker, aplicațiile pot fi portabile și consistente, indiferent de mediul în care sunt rulate – dezvoltare, testare sau producție. Aceasta face din Docker o alegere ideală pentru DevOps, microservicii și scalabilitate.

- Modalități de accesare
 - Local prin instalarea aplicației de pe <https://www.docker.com/products/docker-desktop/> sau prin metode de scripting prin brew pentru mac chocolatey pentru powershell sau instalare de pachete pentru linux
 - Cloud la <https://training.play-with-docker.com/ops-s1-hello/>

Exemple de comenzi docker

- Comenzi utile system
 - pentru afișarea versiunii SO din varianta docker cloud putem rula:

- cat /etc/os-release ce are ca output

```
$ cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.3
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
```

- uname -s ce are ca rezultat

```
[node1] (local) root@192.168.0.13 ~
$ uname -s
Linux node1 4.4.0-210-generic #242-Ubuntu SMP Fri Apr 16 09:57:56 UTC 2021 x86_64 Linux
```

- comenzi generale de docker

- docker -v pentru aflarea versiunii de docker

```
PS C:\Users\bogda> docker -v
Docker version 24.0.6, build ed223bc
```

- pentru listingul de argumente pentru docker `docker --help`

```
PS C:\Users\bogda> docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run          Create and run a new container from an image
  exec         Execute a command in a running container
  ps           List containers
  build         Build an image from a Dockerfile
  pull         Download an image from a registry
  push         Upload an image to a registry
  images       List images
  login         Log in to a registry
  logout        Log out from a registry
  search        Search Docker Hub for images
  version       Show the Docker version information
  info          Display system-wide information
```

RULARE COMENZI DOCKER

- RULARE DOCKER

- Rulare imagine de docker hello-world varianta cloud

- se aduce imaginea cu `docker pull hello-world`

```
[node1] (local) root@192.168.0.13 ~
$ docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
e6590344b1a5: Pull complete
Digest: sha256:bfb0cc14f13f9ed1ae86abc2b9f11181dc50d779807ed3a3c5e55a6936dbdd5
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
```

- se listeaza imaginile cu `docker image ls`

```
$ docker image ls
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
hello-world	latest	74cc54e27dc4	6 weeks ago	10.1kB

- se executa imaginea docker run hello-world (numele imaginii)

```
$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

- Rulare imagine de docker hello-world varianta desktop

- se aduce imaginea cu docker pull hello-world

```
PS C:\Users\bogda> docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
e6590344b1a5: Download complete
Digest: sha256:bfb0cc14f13f9ed1ae86abc2b9f11181dc50d779807ed3a3c5e55a6936dbdd5
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
```

- se listeaza imaginile cu docker image ls

```
PS C:\Users\bogda> docker image ls
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
hello-world         latest          bfb0cc14f13    6 weeks ago    20.4kB
```

- ```
Using default tag: latest
latest: Pulling from library/hello-world
e6590344b1a5: Download complete
Digest: sha256:bfb0cc14f13f9ed1ae86abc2b9f11181dc50d779807ed3a3c5e55a6936dbdd5
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest

What's next:
 View a summary of image vulnerabilities and recommendations → docker scout quickview hello-world
PS C:\Users\bogda> docker image ls
REPOSITORY TAG IMAGE ID CREATED SIZE
hello-world latest bfb0cc14f13 6 weeks ago 20.4kB
PS C:\Users\bogda> docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
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```

- comenzi (docker pull hairyhenderson/figlet, docker image ls, docker run hairyhenderson/figlet Butoi Bogdan Ionut)

```
PS C:\Users\bogda> docker pull hairyhenderson/figlet
Using default tag: latest
latest: Pulling from hairyhenderson/figlet
4b63e4c4693b: Download complete
82401c681f84: Download complete
f18232174bc9: Download complete
Digest: sha256:f2636fedf0c2596a9d6b2e9d86ebd4c7143dc98f6d8dad0fe0de1209b4500c41
Status: Downloaded newer image for hairyhenderson/figlet:latest
docker.io/hairyhenderson/figlet:latest
```

What's next:

View a summary of image vulnerabilities and recommendations → `docker scout quickview hairyhenderson/figlet`

```
PS C:\Users\bogda> docker image ls
```

| REPOSITORY            | TAG    | IMAGE ID     | CREATED     | SIZE   |
|-----------------------|--------|--------------|-------------|--------|
| hairyhenderson/figlet | latest | f2636fedf0c2 | 2 days ago  | 13.2MB |
| hello-world           | latest | bfb0cc14f13  | 6 weeks ago | 20.4kB |

```
PS C:\Users\bogda> docker run hairyhenderson/figlet Butoi Bogdan Ionut
```

- ```
import docker

# Initialize Docker client
client = docker.from_env()
```

```

output = client.containers.run("hairyhenderson/figlet", "salut UTM 2024", remove=True)

output = output.decode("utf-8")

print(output)

```

- Rulare docker din python hello-world

```

import docker

client = docker.from_env()
container = client.containers.run("hello-world")
print(container.decode(" utf-8"))

```

- Rulare docker prin Dockerfile Se genereaza un fisier Dockerfile al carui continut rulat de catre Docker comanda de build a imaginii de docker este **docker build -t xxx -f yyyy** unde xxx reprezinta numele imaginii ce va fi generata iar yyyy reprezinta calea inclusiv numele fisierului Dockerfile
 - Afisarea unui text in terminal Descrierea continutului

```

FROM alpine:latest

CMD ["echo", " WELCOME BUTOI BOGDAN"]

```

From alpine:latest reprezinta ultima versiune a imaginii de linux alpine CMD executa o comanda de linux Pentru a rula imaginea build-uita se foloseste comanda **docker run xxx**

- Afisare textului hello docker/podman world Butoi Bogdan in browser la adresa localhost:50205

```

FROM nginx:latest

WORKDIR /usr/share/nginx/html

COPY index.html index.html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

```

Descriere continutului FROM nginx:latest aduce ultima imagine de nginx WORKDIR /usr/share/nginx/html seteaza calea de randare a fisierelor html COPY index.html index.html copiaza continutul local al fisierului index.html in index.html din /usr/share/nginx/html (WORKDIR) EXPOSE 80 deschide portul 80 de pe container CMD ["nginx", "-g", "daemon off;"] porneste serviciul de nginx Pentru build se pot folosi comenzile de mai sus insa pentru run folosim **docker run -d -p 5205:80 my-nginx** unde my-nginx reprezinta numele containerului -d ruleaza in mod detached (executa si inchide cli-ul) -p 50205:80 mapeaza portul 80 de pe container cu portul 5205 de pe statia de lucru